

CariSurg MedTech Pathways Research Project

Project Title: Development of an Interpretable Digital Triage System for Dynamic Risk Assessment in Emergency Care

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Statement of the Problem:

Emergency department overcrowding delays care for critically ill patients because traditional triage relies on static, point-in-time assessments that fail to capture real-time physiological deterioration in the waiting room. Current AI triage solutions operate as uninterpretable "black boxes," hindering clinical adoption and alignment with established hospital protocols. This 12-week pilot introduces a transparent digital triaging model that translates real-time patient data into actionable risk categories using recognized NEWS2 and SIRS frameworks, safely accelerating clinical throughput and mitigating under-triage errors.

Previous Work:

Extensive peer-reviewed literature establishes that artificial intelligence can significantly optimize emergency department workflows. Recent scoping and narrative reviews demonstrate that machine learning and natural language processing (NLP) models consistently match or exceed human triage accuracy, lowering under-triage rates and accurately predicting secondary clinical outcomes like ICU admission and 24-hour mortality (Stewart et al., 2023; Tyler et al., 2024). Furthermore, advanced multimodal frameworks like TriageIntelli prove that fusing heterogeneous clinical data streams, such as structured vital signs and unstructured text narratives, yields superior predictive accuracy compared to single-modality baselines (Araouchi & Adda, 2024).

However, significant operational boundaries persist in existing models. Most current AI triage systems have only been tested on old hospital records, looking at a single snapshot of data taken the exact moment the patient walked through the front door; as a result, these models completely ignore subsequent physiological shifts and patient degradation during prolonged ED wait times (Sitthiprawiat et al., 2025). Additionally, the clinical integration of automated decision support systems remains severely bottlenecked by the "black-box" problem, where a lack of interpretable feature weighting fuels clinical skepticism and prevents alignment with standardized hospital protocols (Da'Costa et al., 2025). This pilot directly builds upon these gaps by transitioning from static modeling to a transparent, protocol-aligned dynamic framework suitable for real-time deployment.

Literature Review:

Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Lee, D. W., Ginsberg, B., Ahmad, H., & Jacobs, R. J. (2024). Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. *Cureus*, 16(5), e59906.

Clinical/Engineering Problem: Emergency department (ED) triage lacks precision and certainty regarding its overall impact when using traditional systems. There is a critical need to improve earlier diagnosis, patient risk prioritization, resource allocation (like predicting length of stay), and determining which patients accurately require hospitalization.

Method Used: A systemized literature review was conducted in September 2023 across the EMBASE, Ovid MEDLINE, and Web of Science databases.

Reported Outcome: Machine learning models consistently showed superior discrimination and predictive accuracy compared to traditional triage systems. The algorithms successfully enhanced disease identification, accurately determined when urgent patients needed hospital admission, and improved hospital resource allocation and overall quality of care.

Stated Limitation: The text notes that there is still little known overall about the use of AI in ED triage, and it highlights a specific danger where AI algorithms can give "overconfident answers" that could pose safety risks to patients.

Da'Costa, A., Teke, J., Origbo, J. E., Osonuga, A., Egbon, E., & Olawade, D. B. (2025).

AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *International Journal of Medical Informatics*, 197, 105838.

Clinical/Engineering Problem: Emergency Departments (EDs) face severe operational pressure from overcrowding, resource constraints, and traditional triage systems that rely on subjective assessments, leading to a lack of consistency in patient prioritization during peak hours or mass casualty events.

Method Used: A narrative review was conducted analyzing peer-reviewed articles published between 2015 and 2024. The papers were identified through targeted searches across PubMed, Scopus, IEEE Xplore, and Google Scholar databases, with the findings synthesized into a comprehensive overview.

Reported Outcome: The synthesized findings showed substantial benefits for AI-driven triage, including improved patient prioritization, reduced wait times, and optimized hospital resource allocation, demonstrating a strong potential to transform ED operations and improve patient outcomes in high-pressure environments.

Stated Limitation: Widespread adoption is significantly blocked by operational and technical challenges, specifically data quality issues, algorithmic bias, a lack of clinician trust, and unresolved ethical concerns regarding equitable implementation.

Araouchi, Z., & Adda, M. (2024). TriageIntelli: AI-Assisted multimodal triage system for health centers. *Procedia Computer Science*, 251, 430–437.

Clinical/Engineering Problem: Traditional emergency department triage methods—like the Emergency Severity Index (ESI) and the Canadian Triage and Acuity Scale (CTAS)—suffer from human subjectivity, cannot handle severe overcrowding efficiently, and lead to poor hospital resource allocation.

Method Used: A literature review examining and comparing the evolutionary transition of emergency triage systems from conventional, manual protocols over to modern AI-assisted predictive models.

Reported Outcome: The review found that AI models offer enhanced diagnostic precision, optimized patient prioritization, and a reduction in human error, which collectively streamlines busy emergency room workflows.

Stated Limitation: The implementation of these models faces critical bottlenecks, specifically heavy data dependency, unresolved ethical challenges, and inconsistent, variable performance when deployed across different healthcare settings.

Sitthiprawiat, P., Wittayachamnankul, B., Sirikul, W., & Laohavisudhi, K. (2025).

Development and internal validation of an AI-based emergency triage model for predicting critical outcomes in emergency department. *Scientific Reports*, 15(1), 31212.

Clinical/Engineering Problem: Severe emergency department (ED) overcrowding delays patient care and worsens clinical outcomes because traditional triage systems have inherent limitations in accuracy and consistency when identifying who needs urgent resources.

Method Used: A retrospective analysis of 163,452 ED visits (from 2018 to 2022) compared logistic regression, random forest, and XGBoost machine learning models against the traditional Canadian Triage and Acuity Scale (CTAS). The models evaluated mode of arrival, patient age, and free-text chief complaints using advanced multilingual sentence embeddings.

Reported Outcome: The XGBoost model achieved superior predictive performance for ICU admission and resource needs over traditional CTAS (AUROC 0.917 vs. 0.882).

Incorporating unstructured text data via machine learning significantly enhanced triage accuracy and resource allocation by identifying critically ill patients more effectively.

Stated Limitation: The study relied on a retrospective analysis of historical data from a single facility (Maharaj Nakhon Chiang Mai Hospital), meaning the model was validated internally and its performance may vary when deployed in a live, real-time clinical environment.

Stewart, J., Lu, J., Goudie, A., Arendts, G., Meka, S. A., Freeman, S., Walker, K., Sprivulis, P., Sanfilippo, F., Bennamoun, M., & Dwivedi, G. (2023). Applications of natural language processing at emergency department triage: A narrative review. *PLoS ONE*, 18(12), e0279953.

Clinical/Engineering Problem: Standard emergency triage risk stratification techniques struggle to predict clinical outcomes like hospital admission and critical illness because they completely ignore the rich, unstructured human language contained within free-text nursing notes.

Method Used: A scoping review and narrative synthesis screened 3,730 studies across MEDLINE, Embase, Cochrane, Web of Science, and Scopus to evaluate 20 peer-reviewed papers that applied NLP to free-text data acquired at ED triage. The researchers used the Prediction model Risk of Bias Assessment Tool (PROBAST) to systematically evaluate the quality and bias of the included predictive models.

Reported Outcome: NLP models achieved high accuracy in predicting triage scores, hospital admissions, and critical illness. Furthermore, combining free-text data with structured data consistently improved model predictive performance compared to using structured data alone.

Stated Limitation: The findings are heavily restricted because 80% of the reviewed studies carry a high risk of bias, almost all the research is retrospective, and there is a critical shortage of prospective clinical trials or real-world model deployment in live practice.

Proposed Solution:

To fix the problems of rigid risk scoring and the lack of trust in confusing healthcare AI, this project proposes a clear digital triaging tool for real-time decision support in the Mercer General Hospital Emergency Department. Instead of giving a confusing percentage based only on a single snapshot from when the patient first arrives, this software will update using new patient data as it comes in. The system will look at both the standard numbers like vital signs and the notes written by nurses about the patient's main complaints, which recent studies show helps make much better predictions.

Crucially, the system builds clinical trust by connecting its findings directly to standard medical scoring systems that staff already know, specifically the NEWS2 and SIRS frameworks. If a patient gets worse while sitting in an overcrowded waiting room, the system will automatically recalculate their risk level. It will then display a quick explanation of why their priority has gone up, such as a dangerous change in their breathing or heart rate.

Over a 12 week pilot window, this tool will run quietly in the background to shadow Mercer's current triage workflows without interrupting daily care. By turning complex computer calculations into familiar risk categories, this solution will help frontline staff quickly spot fading patients, speed up emergency room wait times safely, and prevent dangerous under-triaging errors.

Boulitsakis Logothetis, S., Green, D., Holland, M., & Al Moubayed, N. (2023). Predicting acute clinical deterioration with interpretable machine learning to support emergency care decision making. *Scientific reports*, 13(1), 13563.
<https://doi.org/10.1038/s41598-023-40661-0>

Clinical/Engineering Problem: Increasing operational pressures in fast-paced Emergency Departments create an urgent need to efficiently identify patients at risk of rapid health decline. Traditional tracking frameworks like NEWS2 can cause alert fatigue or entirely miss high-risk patients who present with low initial scores.

Method Used: A retrospective study analyzing a dataset of 118,886 unplanned hospital admissions at Salford Royal Hospital, UK. The researchers systematically compared machine learning algorithms (logistic regression, gradient-boosted decision trees, and support vector machines) using a patient's first recorded vital signs, lab results, and observations to predict mortality or critical care needs within 24 hours. The team used Shapley Additive exPlanations (SHAP) to map out feature weights and justify model outputs.

Reported Outcome: The machine learning models significantly outperformed the traditional NEWS2 framework, yielding up to a 0.366 increase in average precision, a notable reduction

in daily false alarms (alert fatigue), and a drop in biased predictions across different age and sex demographics.

Stated Limitation: The models were built on historical, cross-sectional data from a single hospital site, and the authors explicitly noted that further work is still needed to trial and test these models in actual, live clinical practice.

What it Proves: This study directly evaluates machine learning algorithms (such as gradient-boosted decision trees) against the traditional NEWS2 framework using Electronic Health Record (EHR) data from ED patients.

Liu, N., Xie, F., Siddiqui, F. J., Ho, A. F. W., Chakraborty, B., Nadarajan, G. D., Tan, K. B. K., & Ong, M. E. H. (2022). Leveraging Large-Scale Electronic Health Records and Interpretable Machine Learning for Clinical Decision Making at the Emergency Department: Protocol for System Development and Validation. *JMIR research protocols*, 11(3), e34201. <https://doi.org/10.2196/34201>

Clinical/Engineering Problem: Growing global demand and overcrowding in Emergency Departments lead to longer waiting times. While machine learning tools can build triage and risk prediction models to help, the "black box" nature of these advanced models limits their actual clinical use and interpretation by staff.

Method Used: A retrospective, single-center study protocol analyzing a massive longitudinal dataset of approximately 1.8 million ED visits (representing over 810,000 unique patients) from a tertiary hospital in Singapore. The researchers used a machine learning tool called *AutoScore* to create three dynamic, interpretable scores (called SERT) to evaluate patients at different times: on arrival, during their ED stay, and at admission.

Reported Outcome: The system successfully creates a unique scoring system that is both dynamic and transparent. It establishes a clear standard for using large-scale Electronic Health Records (EHRs) and interpretable machine learning to predict severe outcomes, like the need for an intensive care unit (ICU) or inpatient death.

Stated Limitation: The paper describes a retrospective study protocol and system design rather than a finished prospective clinical trial, meaning the models still require further real-world validation to prove their clinical safety and impact in live practice.

What it Proves: The authors outline the clinical imperative for moving away from 5-level static tools (like the Emergency Severity Index) and standard scores because they are subjective, static, and fail to track patients whose conditions rapidly change or deteriorate during an ED stay.

References:

Araouchi, Z., & Adda, M. (2024). TriageIntelli: AI-Assisted multimodal triage system for health centers. *Procedia Computer Science*, 251, 430–437. <https://doi.org/10.1016/j.procs.2024.11.130>

Boulitsakis Logothetis, S., Green, D., Holland, M., & Al Moubayed, N. (2023). Predicting acute clinical deterioration with interpretable machine learning to support emergency care decision making. *Scientific reports*, 13(1), 13563. <https://doi.org/10.1038/s41598-023-40661-0>

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